Xinyu Ren

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EDUCATION

Cornell University, Ithaca, NY 09/2022-12/2025

- *B.S. Information Science; GPA: 3.86/4.3; Minor in Business; GPA: 3.68/4.3*
- Dean's List (05/2025, 12/2024, 05/2024, 12/2023, 05/2023, 12/2022)

SKILLS & TOOLS

- UX Research: User Interviews, Usability Testing, Survey Design, Persona Creation, Affinity Diagramming, Qualitative & Quantitative Data Analysis
- Programming: Python, Java, HTML, CSS, JavaScript, R, Bootstrap, PHP
- Design: Figma, FigJam, Miro
- Language: Chinese (Native), English (Fluent), Korean (Beginner), Spanish (Beginner)

LAB EXPERIENCE

Architectural Robotics Lab (ARL) 07/2024-Present

Research Assistant, Tutor: Prof. Keith Evan Green

- Participate in the design and development of robotic systems that augment and support human activities in constrained and dynamic environments, including building physical prototypes, conducting literature reviews, synthesizing research findings, designing study protocols, and collecting and coding user study data.

Projects included:

- ***RobotRoom***: https://www.serenageguo.com/robotroom 01/2025-Present
  a physical reconfigurable foldable robotic system that affords multiple activities for people living in limited space.
- Conduct literature reviews in Human-Robot Interaction and adaptive environments to inform system design.
- Participate in ideation of interaction modality of reconfigurable smart home, developing design concepts through scenario mapping and interface sketching.
- Design user survey and performed quantitative and qualitative data analysis on 200+ responses to evaluate usability and user perception.
- Supported physical prototyping and fabrication in a workshop setting.
- ***MirrorBot***: https://www.serenageguo.com/mirrorbot 01/2025-08/2025
  a robotic mirror that promotes ice-breaking and fosters serendipitous human connections.
- Conducted literature reviews on the historical, commercial, and psychological uses of mirrors in interaction design and behavior shaping.
- Designed and piloted a user study protocol, leading participant recruitment and facilitating 20+ experimental sessions and interviews.
- Coded interviews and performed qualitative coding of video data to identify behavioral patterns and social dynamics.
- Developed a structured codebook to synthesize insights and illustrative quotes, directly supporting paper writing.
- **DentAR**: https://www.serenageguo.com/dentar 07/2024-12/2024
  an AR sensory modulation experience for people with heightened sensory sensitivity toward the dental clinic's physical environment.
- Conducted literature reviews on autism/sensory processing disorder, participated in pilot co-design sessions to provide feedback, and promoted the study through campus poster campaigns.
- Coded qualitative data from co-design sessions and interviews and contributed to developing a codebook.
- Analyzed sketches and co-designed outputs collected from the study sessions to synthesize emotional needs.

Design + Augmented Intelligence Lab (DAIL) 04/2024 - 05/2025

Research Assistant, Tutor: Prof. Saleh Kalantari

- Conducted literature reviews on 35+ papers and contributed to the syllabus of the Human-AI Interaction (INFO 3450).
- Assisted in research, data collection, and analysis for VR projects on human-technology interaction.

Projects included:

- **Intersection** 04/2024-05/2025
  a project that utilized VR to assist elder people to navigate through an unfamiliar environment.
- Invited participants to play a VR treasure hunt game for usability study and processed data from 140+ participants in collaboration with a 4-member team.
- **LocoMotion** 09/2024-05/2025
  a project that utilized VR to make hospitals easier to navigate for all
- Contributed to promotion of the study and collected and processed 50+ data.

PRACTICAL EXPERIENCE

Cornell University Bowers College of Computing and Information Science 08/2025-Present

Teaching Assistant for INFO 3450 (Human-Computer Interaction) Class Size: 157

- Mentor students through the full human-centered design cycle, providing constructive feedback on their project proposals and prototypes to strengthen their design thinking.
- Assess students' assignments to support studying outcomes and maintain academic integrity.

Shanshiping Village Online Museum Project 07/2025-Present

Website Designer

- Co-work with an international high school in Shanghai and Yunnan Red Cross Society to design an official museum website for the leprosy villages in Yunnan Province.
- Provide suggestions on content strategy and information architecture, outlined a clear framework for content collection and site structure, and led end-to-end website design and development.
- Manage ongoing website maintenance and updates to ensure optimal user experience post-launch.

Museum Link: https://www.shanshiping.com/english

Field Research on Retiree Migrants in Fangchenggang, Guangxi Guangxi, China/Remote

Independent Researcher 01/2025-Present

Research aims to inform the design of simple, accessible tools to help older migrants better navigate their new communities.

- Conduct an in-depth field study on the experiences of 20+ middle-aged and elderly retirees relocating to Fangchenggang, a growing winter destination in southern China.
- Investigate challenges that older adults face in accessing local community resources (e.g., hospitals, parks, tourist sites) due to limited digital literacy.
- Conduct remote interviews, digital ethnography (WeChat forums, local content platforms), and online observations to understand user pain points for the future walk-along interviews for on-site fieldwork in Fangchenggang.

Guangxi Henghua Automobile Co., Ltd. Remote

UX Research & Strategy Intern 07/2025-09/2025

- Conducted a strategic UX analysis of 5+ automotive service mini-programs and finished competitive analysis.
- Delivered analytical reports and prioritized design proposals for future strategies.

Community Reconstruction Research Project Guangxi, China

Research Leader 06/2021-06/2022

- Led multiple field surveys in Shanglin County of Guangxi to find methods of promoting rural tourism, collective public governance and cultural inheritance in ancient villages through the protection of cultural elements.

COURSE PROJECT

Mobility4All - Equitable Transportation Project-*INFO 4420 Human Computer Interaction Studio Project*

Project Leader 08/2025-Present

- Initiated the discovery phase for an equitable mobility project, co-developing the problem space with teammates through comprehensive literature review and design space analysis to establish a foundational research base.
- Conducted and synthesized 10+ stakeholder interviews (including students, low-income residents, individuals with disabilities, transportation operators, and researchers in the field) to uncover critical barriers to reliable transportation.
- Led the team to translate multi-perspective user and stakeholder insights into clear design opportunities through facilitated synthesis sessions, effectively communicating research findings and design implications in presentations.

FitMatch - Online Shopping Experience Empowerment Project-*INFO 4400 Qualitative User Research&Design Method*

Project Leader 01/2025-05/2025

- Led a 4-person team through full-cycle qualitative research on female college students' online shopping experiences.
- Synthesized qualitative research findings(ethnography, interviews, and co-design) into actionable design principles.
- Delivered a high-fidelity prototype, a research poster, and a video demonstration.

BrainBud - Attention Tracking App Project- *INFO 3450 Human-Computer Interaction*

Project Leader 08/2024-12/2024

- Led a 4-member team in designing a mobile app that helps students with ADHD maintain focus during lectures.
- Developed wireframes and interaction prototype in Figma, and conducted usability tests to identify key pain points.
- Iterated on designs based on heuristic evaluation and user feedback, delivering a high-fidelity prototype accompanied by a research poster and video demonstration.

Website Design Project-*INFO 2300 Design and Programming for Web*

Project Leader 10/2023-12/2023

- Led a 5-member team to help build a personal website for Dr. Dhaniyala, a professor of the Department of Mechanical & Aerospace Engineering of Clarkson University.
- Constructed basic web structure using HTML, styling pages using CSS and implementing interactivity using JavaScript.
- Organized weekly meetings with team members, ensuring effective communication and providing consistent updates on project progress to the client.

Sewer Network Maze Project - *CS 2110 Object-Oriented Programming and Data Structures*

Project Team Member 11/2023-12/2023

- Collaborated with a classmate to complete a Sewer network maze game using Java.
- Synthesized different content and skills, including Dijkstra's algorithm, synchronization, and GUI.